# Bellamkonda Ankitha

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2028 | 7.32 |
| Sri Chaitanya Jr Kalasala | Class XII (Board of Intermediate Education) | 2024 | 81.00% |
| Sri Chaitanya High School | Class X (Board of Secondary Education) | 2022 | 9.70% |

## SCHOLASTIC ACHIEVEMENTS

Secured **Grade A in Microbiology, Biochemistry, and Nucleic Acids and RNA Biomedicine**, along with **Grade A in Biological Engineering and Biostatistics**, demonstrating a Better academic foundation in **microbiology, biochemistry, biological engineering, molecular biology, and quantitative analysis**.

## POSITIONS OF RESPONSIBILITY

**VCU Head | Prayas, IIT Hyderabad | 2025-2026**

Led the Volunteer Coordination Unit, managing volunteer recruitment, team coordination, task allocation, and execution of key initiatives.

**Hospitality Coordinator | Prayas, IIT Hyderabad | 2024-2025**

Managed accommodation, hospitality, and logistical arrangements for guests and participants during major Prayas events, ensuring smooth coordination and timely execution.

**Hospitality Coordinator | Inter-IIT Sports Meet, IIT Hyderabad | 2025**

Coordinated accommodation and hospitality arrangements for visiting teams and officials, ensuring efficient communication and smooth on-campus logistics throughout the event.

## SKILLS

**Programming & Analysis:** Python (Pandas, NumPy, Matplotlib), basic Machine Learning and AI concepts, Data Structures, Statistics/Biostatistics, Excel

**Domain Tools:** PyMOL, Solid Edge

**Laboratory & Biotechnology:** Microbiology, Microbial Culture, Streaking Techniques, Biochemistry, Basic Biotechnology Laboratory Techniques, Autoclaving, Biochemical Engineering, Molecular Biology, PCR, Gel Electrophoresis, and SDS-PAGE.

**Bioinformatics & Data Analysis:** Basic Bioinformatics, Next-Generation Sequencing, Biological Databases, Biostatistics, Experimental Data Analysis, R, Python, Pandas, Matplotlib, and Microsoft Excel.

## RELEVANT COURSES

Introduction to Programming, Data Structures & Applications, and Artificial Intelligence -- covering core programming, data structures, and AI fundamentals relevant to technical roles.

Microbiology (A), Biochemistry (A), Nucleic Acids and RNA Biomedicine (A), Biological Engineering (A), Biostatistics (A), Biochemical Engineering, Basic Biotechnology Lab, Basic Bioinformatics, R and Python, and Big Data Biology & Biological Databases.

## PROJECTS

**Microbial Culture and Isolation**

Performed **microbial culture and streaking techniques** to obtain isolated colonies from microbial samples. Practiced aseptic handling and proper laboratory procedures to minimize contamination. Used **autoclaving and sterilization techniques** for media and laboratory equipment and maintained proper experimental conditions.

**PCR and Agarose Gel Electrophoresis**

Performed **Polymerase Chain Reaction (PCR)** for amplification of target DNA sequences and analyzed amplified products using **agarose gel electrophoresis**. Prepared samples, handled DNA, and interpreted band patterns to evaluate experimental results.

**Protein Separation and Analysis using SDS-PAGE**

Performed **SDS-PAGE (Sodium Dodecyl Sulfate-Polyacrylamide Gel Electrophoresis)** to separate proteins based on molecular size. Prepared protein samples and gels, carried out electrophoresis, and analyzed the resulting protein band patterns.

**P vs NP in Genomic Data Security**

Explored the **P vs NP problem** in the context of RSA encryption for securing genomic data. Implemented prime factorization in Python for composite numbers up to 10 digits, analyzed runtime growth with increasing input size, and visualized computational performance using **NumPy and Matplotlib**. Discussed the potential impact of quantum computing on current encryption methods.

**Tools:** Python, NumPy, Matplotlib | **Course:** Biostatistics